Book: Mechatronics An Integrated Approach by Clarence W. De Silva

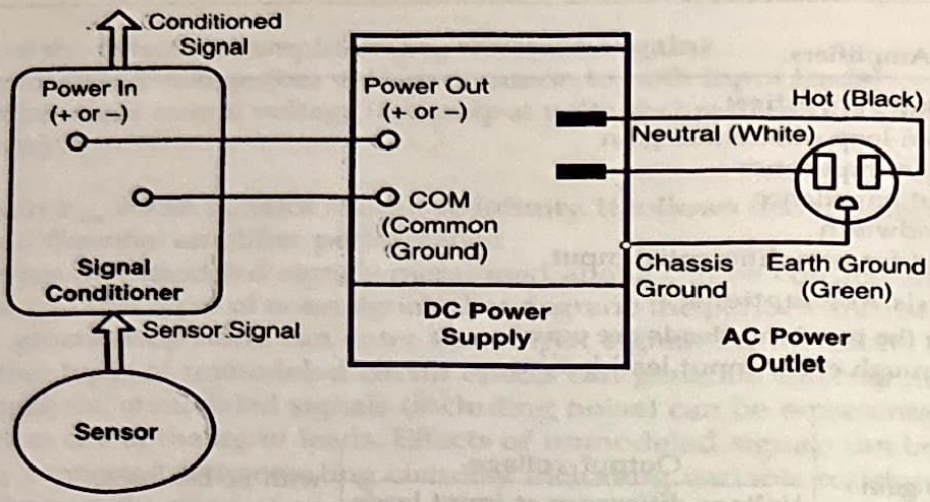


FIGURE 4.10
An example of grounding of instruments.

to the signal ground of the signal conditioning module. In this manner a common 0 V reference is provided for the dc voltage that is supplied to the signal conditioning module.

A main cause of electrical noise is the ground loops, which are created due to improper grounding of instruments. If two interconnected instruments are grounded at two separate locations that are far apart (multiple grounding), ground loop noise can enter the signal leads because of the possible potential difference between the two ground points. The reason is that ground itself is not generally a uniform-potential medium, and a nonzero (and finite) impedance may exist from point to point within this medium. This is, in fact, the case with a typical ground medium such as a common ground wire. An example is shown schematically in Figure 4.11(a). In this example, the two leads of a sensor are directly connected to a signal-conditioning device such as an amplifier, one of its input leads (+) being grounded (at point B). The 0 V reference lead of the sensor is grounded through its housing to the earth ground (at point A). Because of nonuniform ground potentials, the two ground points A and B are subjected to a potential difference v_g. This will create a ground loop with the common reference lead, which interconnects the two devices. The solution to this problem is to isolate (i.e., provide an infinite impedance to) either one of the two devices. Figure 4.11b shows internal isolation of the sensor. External isolation, by insulating the housing of the sensor, will also remove the ground loop. Floating off the common ground (COM) of a power supply (see Figure 4.10) is another approach to eliminating ground loops. Specifically, COM is not connected to earth ground.

4.4 Analog Filters

A filter is a device that allows through only the desirable part of a signal, rejecting the unwanted part. Unwanted signals can seriously degrade the performance of a mechatronic system. External disturbances, error components in excitations, and noise generated internally within system components and instrumentation are such spurious signals, which may be removed by a filter. As well, a filter is capable of shaping a signal in a desired manner.

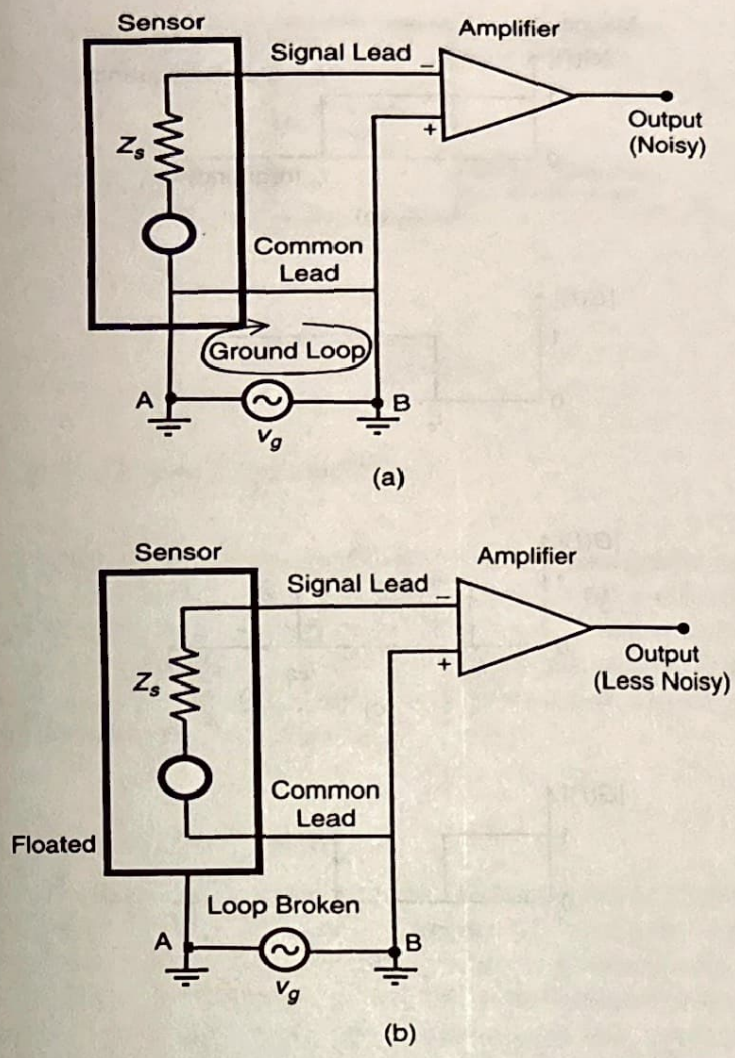


FIGURE 4.11
(a) Illustration of a ground loop, (b) Device isolation to eliminate ground loops (an example of internal isolation).

In typical applications of acquisition and processing of signals in a mechatronic system, the filtering task would involve the removal of signal components in a specific frequency range. In this context we can identify the following four broad categories of filters:

1. Low-pass filters
2. High-pass filters
3. Band-pass filters
4. Band-reject (or notch) filters

The ideal frequency-response characteristic of each of these four types of filters is shown in Figure 4.12. Note that only the magnitude of the frequency response function (magnitude of the frequency transfer function) is shown. It is understood, however, that the phase distortion of the input signal also should be small within the *pass band* (the allowed frequency range). Practical filters are less than ideal. Their frequency response functions do not exhibit sharp cutoffs as in Figure 4.12 and, furthermore, some phase distortion will be unavoidable.

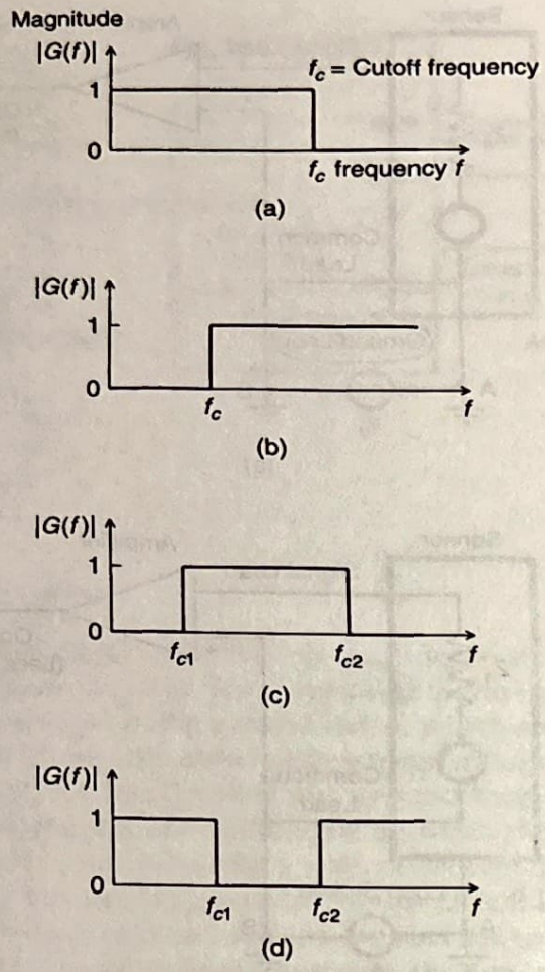


FIGURE 4.12
Ideal filter characteristics: (a) Low-pass filter, (b) High-pass filter, (c) Band-pass filter, (d) Band-reject (notch) filter.

A special type of band-pass filter that is widely used in acquisition and monitoring of response signals (e.g., in product dynamic testing) is *tracking filter*. This is simply a band-pass filter with a narrow pass band that is frequency-tunable. The center frequency (mid value) of the pass band is variable, usually by coupling it to the frequency of a carrier signal (e.g., drive signal). In this manner, signals whose frequency varies with some basic variable in the system (e.g., rotor speed, frequency of a harmonic excitation signal, frequency of a sweep oscillator) can be accurately tracked in the presence of noise. The inputs to a tracking filter are the signal that is being tracked and the variable *tracking frequency* (*carrier input*). A typical tracking filter that can simultaneously track two signals is schematically shown in Figure 4.13.

Filtering can be achieved by *digital filters* as well as *analog filters*. Before digital signal processing became efficient and economical, analog filters were exclusively used for signal filtering, and are still widely used. An analog filter is typically an *active filter* containing active components such as transistors or op-amps. In an analog filter, the input signal is passed through an analog circuit. Dynamics of the circuit will determine which (desired) signal components would be allowed through and which (unwanted) signal components would be rejected. Earlier versions of analog filters employed discrete circuit elements such as discrete transistors, capacitors, resistors and even discrete inductors. Since inductors

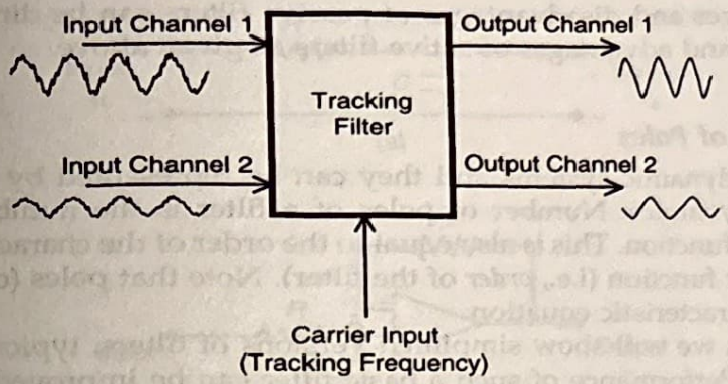


FIGURE 4.13
Schematic representation of a two-channel tracking filter.

have several shortcomings such as susceptibility to electromagnetic noise, unknown resistance effects, and large size, today they are rarely used in filter circuits. Furthermore, due to well-known advantages of integrated circuit (IC) devices, today analog filters in the form of monolithic IC chips are extensively used in mechatronic applications and are preferred over discrete-element filters. Digital filters, which employ digital signal processing to achieve filtering, are also widely used today.

4.4.1 Passive Filters and Active Filters

Passive analog filters employ analog circuits containing passive elements such as resistors and capacitors (and sometimes inductors) only. An external power source is not needed in a passive filter. Active analog filters employ active elements and components such as transistors and operational amplifiers in addition to passive elements. Since external power is needed for the operation of the active elements and components, an active filter is characterized by the need of an external power supply. Active filters are widely available in a monolithic integrated-circuit (IC) package and are usually preferred over passive filters.

Advantages of active filters include the following:

1. Loading effects and interaction with other components are negligible because active filters can provide a very high input impedance and a very low output impedance.
2. They can be used with low signal levels because both signal amplification and filtering can be provided by the same active circuit.
3. They are widely available in a low-cost and compact integrated-circuit form.
4. They can be easily integrated with digital devices.
5. They are less susceptible to noise from electromagnetic interference.

Commonly mentioned disadvantages of active filters are the following:

1. They need an external power supply.
2. They are susceptible to "saturation" type nonlinearity at high signal levels.
3. They can introduce many types of internal noise and unmodeled signal errors (offset, bias signals, etc.).

Note that advantages and disadvantages of passive filters can be directly inferred from the disadvantages and advantages of active filters as given above.

4.4.1.1 Number of Poles

Analog filters are dynamic systems and they can be represented by transfer functions, assuming linear dynamics. Number of poles of a filter is the number of poles in the associated transfer function. This is also equal to the order of the characteristic polynomial of the filter transfer function (i.e., *order* of the filter). Note that poles (or, eigenvalues) are the roots of the characteristic equation.

In our discussion we will show simplified versions of filters, typically consisting of a single filter stage. Performance of such a basic filter can be improved at the expense of circuit complexity (and increased pole count). Only simple discrete-element circuits are shown for passive filters. Basic operational-amplifier circuits are given for active filters. Even here, much more complex devices are commercially available, but our purpose is to illustrate the underlying principles rather than to provide complete descriptions and data sheets for commercial filters.

4.4.2 Low-Pass Filters

The purpose of a low-pass filter is to allow through all signal components below a certain (cutoff) frequency and block off all signal components above that cutoff. Analog low-pass fitters are widely used as *antialiasing filters* in digital signal processing. An error known as aliasing will enter the digitally processed results of a signal if the original signal has frequency components above half the *sampling frequency* (half the sampling frequency is called the *Nyquist frequency*). Hence, aliasing distortion can be eliminated if the signal is filtered using a low-pass filter with its cutoff set at Nyquist frequency, prior to sampling and digital processing (See Chapter 5). This is one of numerous applications of analog low-pass filters. Another typical application would be to eliminate high-frequency noise in a measured system response.

A single-pole passive low-pass filter circuit is shown in Figure 4.14(a). An active filter corresponding to the same low-pass filter is shown in Figure 4.14(b). It can be shown that the two circuits have identical transfer functions. Hence, it might seem that the op-amp in Figure 4.14(b) is redundant. This is not true, however. If two passive filter stages, each similar to Figure 4.14(a) are connected together, the overall transfer function is not equal to the product of the transfer functions of the individual stages. The reason for this apparent ambiguity is the circuit loading (interaction) that arises due to the fact that the input impedance of the second stage is not sufficiently larger than the output impedance of the first stage. But, if two active filter stages similar to Figure 4.14(b) are connected together, such loading errors will be negligible because the op-amp with feedback (i.e., a voltage follower) introduces a very high input impedance and very low output impedance, while maintaining the voltage gain at unity. With similar reasoning it can be concluded that an active filter has the desirable property of very low interaction with any other connected component.

To obtain the filter equation for Figure 4.14(a) note that since the output is in open circuit (zero load current), the current through capacitor C is equal to the current through resistor R. Hence,

$$C\frac{dv_o}{dt} = \frac{v_i - v_o}{R}$$

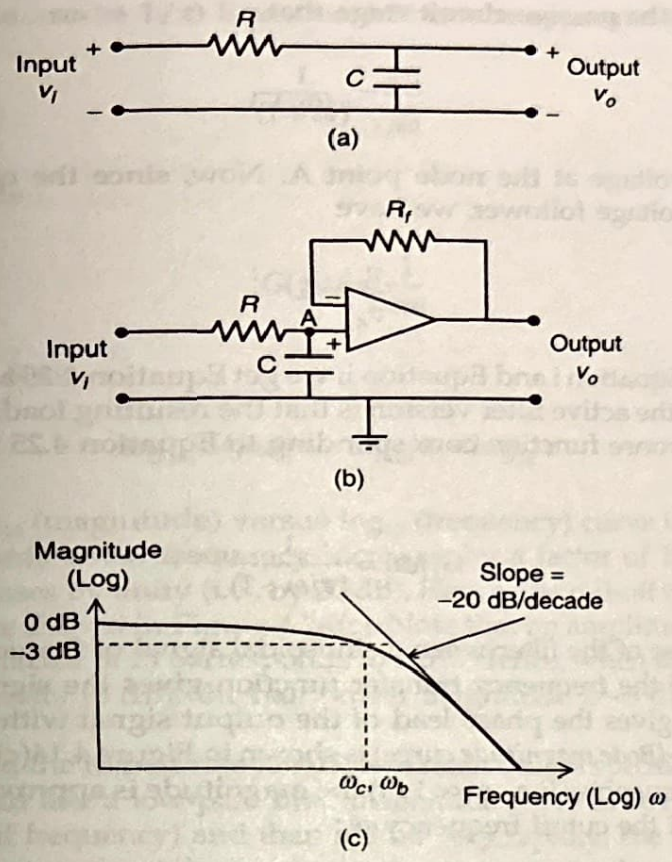


FIGURE 4.14
A single-pole low-pass filter: (a) A passive filter stage, (b) An active filter stage, (c) The frequency response characteristic.

or

$$\tau \frac{dv_o}{dt} + v_o = v_i \tag{4.23}$$

where the filter time constant is

$$\tau = RC \tag{4.24}$$

Now from Equation 4.23 it follows that the filter transfer function is

$$\frac{v_o}{v_i} = G(s) = \frac{1}{(\tau s + 1)} \tag{4.25}$$

From this transfer function it is clear that an analog low-pass filter is essentially a *lag circuit* (i.e., it provides a phase lag).

It can be shown that the active filter stage in Figure 4.14(b) has the same input/output equation. First, since current through an op-amp lead is almost zero, it follows from the

previous analysis of the passive circuit stage that

$$\frac{v_A}{v_i} = \frac{1}{(\tau s + 1)} \tag{i}$$

in which v_A is the voltage at the node point A. Now, since the op-amp with feedback resistor is in fact a voltage follower, we have

$$\frac{v_o}{v_A} = 1 \tag{ii}$$

Next, by combining Equation i and Equation ii we get Equation 4.25 as required. Repeating, a main advantage of the active filter version is that the resulting loading error is negligible.

The frequency response function corresponding to Equation 4.25 is obtained by setting $s = j\omega$; thus

$$G(j\omega) = \frac{1}{(\tau j\omega + 1)} \tag{2.26}$$

This gives the response of the filter when a sinusoidal signal of frequency ω is applied. The magnitude $|G(j\omega)|$ of the frequency transfer function gives the signal amplification and phase angle $\angle G(j\omega)$ gives the phase lead of the output signal with respect to the input. The magnitude curve (*Bode magnitude curve*) is shown in Figure 4.14(c). Note from Equation 4.26 that for small frequencies (i.e., $\omega \ll 1/\tau$) the magnitude is approximately unity. Hence, $1/\tau$ can be considered the cutoff frequency ω_c:

$$\omega_c = \frac{1}{\tau} \tag{4.27}$$

Example 4.6

Show that the cutoff frequency given by Equation 4.27 is also the *half-power bandwidth* for the low-pass filter. Show that for frequencies much larger than this, the filter transfer function on the Bode magnitude plane (i.e., log magnitude versus log frequency) can be approximated by a straight line with slope –20 dB/decade. This slope is known as the *roll-off rate*.

SOLUTION

The frequency corresponding to half power (or $1/\sqrt{2}$ magnitude) is given by

$$\frac{1}{|\tau j\omega + 1|} = \frac{1}{\sqrt{2}}$$

By cross multiplying, squaring, and simplifying the equation we get

$$\tau^2\omega^2 = 1$$

Hence, the half-power bandwidth is

$$\omega_b = \frac{1}{\tau} \tag{2.28}$$

This is identical to the cutoff frequency give by Equation 4.27.

Now for $\omega \gg 1/\tau$ (i.e., $\tau\omega \gg 1/\tau$) Equation 4.26 can be approximated by

$$G(j\omega) = \frac{1}{\tau j\omega}$$

This has the magnitude

$$|G(j\omega)| = \frac{1}{\tau\omega}$$

Converting to the log scale,

$$\log_{10}|G(j\omega)| = -\log_{10}\omega - \log_{10}\tau$$

It follows that the $\log_{10}$ (magnitude) versus $\log_{10}$ (frequency) curve is a straight line with slope –1. In other words when frequency increases by a factor of 10 (i.e., a decade) the $\log_{10}$ magnitude decreases by unity (i.e., by 20 dB). Hence, the roll-off rate is –20 dB/decade. These observations are shown in Figure 4.14(c). Note that an amplitude change by a factor of $\sqrt{2}$ (or power by a factor of 2) corresponds to 3 dB. Hence, when the dc (zero-frequency magnitude) value is unity (0 dB), the half power magnitude is –3 dB.

Cutoff frequency and the roll-off rate are the two main design specifications for a low-pass filter. Ideally, we would like a low-pass filter magnitude curve to be flat up to the required pass-band limit (cutoff frequency) and then roll off very rapidly. The low-pass filter shown in Figure 4.14 only approximately meets these requirements. In particular, the roll-off rate is not large enough. We would prefer a roll-off rate of at least –40 dB/decade and even –60 dB/decade in practical filters. This can be realized by using a high order filter (i.e., a filter having many poles). Low-pass Butterworth filter is of this type and is widely used.

4.4.2.1 *Low-Pass Butterworth Filter*

A low-pass Butterworth filter having 2 poles can provide a roll-off rate of –40 dB/decade and, one having 3 poles can provide a roll-off rate of –60 dB/decade. Furthermore, the steeper the roll-off slope, the flatter the filter magnitude curve within the pass band.

A two-pole, low-pass Butterworth filter is shown in Figure 4.15. We could construct a two-pole filter simply by connecting together two single-pole stages of the type shown in

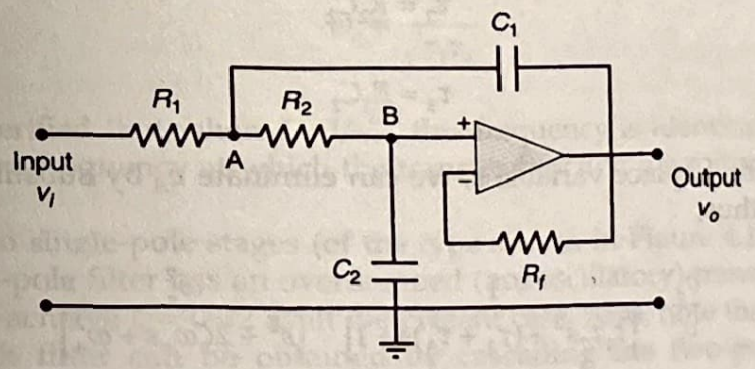


FIGURE 4.15
A two-pole low-pass Butterworth filter.

Figure 4.14(b). Then, we would require two op-amps, whereas the circuit shown in Figure 4.15 achieves the same objective by using only one op-amp (i.e., at a lower cost).

Example 4.7

Show that the op-amp circuit in Figure 4.15 is a low-pass filter having two poles. What is the transfer function of the filter? Estimate the cutoff frequency under suitable conditions. Show that the roll-off rate is –40 dB/decade.

SOLUTION

To obtain the filter equation, we write the current balance equations first. Specifically, the sum of the currents through R_1 and C_1 passes through R_2. The same current has to pass through C_2 because the current through the op-amp lead is zero (a property of an op-amp). Hence,

$$\frac{v_i - v_A}{R_1} + C_1 \frac{d}{dt}(v_o - v_A) = \frac{v_A - v_B}{R_2} = C_2 \frac{dv_B}{dt} \tag{i}$$

Also, since the op-amp with a feedback resistor R_f is a voltage follower (with unity gain), we have

$$v_B = v_o \tag{ii}$$

From Equation i and Equation ii we get

$$\frac{v_i - v_A}{R_1} + C_1 \frac{dv_o}{dt} - C_1 \frac{dv_A}{dt} = C_2 \frac{dv_o}{dt} \tag{iii}$$

$$\frac{v_A - v_o}{R_2} = C_2 \frac{dv_o}{dt} \tag{iv}$$

Now defining the constants

$$\tau_1 = R_1 C_1 \tag{4.29}$$

$$\tau_2 = R_2 C_2 \tag{4.30}$$

$$\tau_3 = R_1 C_2 \tag{4.31}$$

and introducing the Laplace variable s, we can eliminate v_A by substituting Equation iv into Equation iii; thus,

$$\frac{v_o}{v_i} = \frac{1}{\left[\tau_1 \tau_2 s^2 + (\tau_2 + \tau_3)s + 1\right]} = \frac{\omega_n^2}{\left[s^2 + 2\zeta\omega_n s + \omega_n^2\right]} \tag{4.32}$$

This second order transfer function becomes oscillatory if $(\tau_2 + \tau_3)^2 < 4\tau_1\tau_2$. Ideally, we would like to have a zero resonant frequency, which corresponds to a damping ratio value

$\zeta = 1/\sqrt{2}$. Since the undamped natural frequency is

$$\omega_n = \frac{1}{\sqrt{\tau_1 \tau_2}} \tag{4.33}$$

the damping ratio is

$$\zeta = \frac{\tau_2 + \tau_3}{\sqrt{4\tau_1 \tau_2}} \tag{4.34}$$

and the resonant frequency is

$$\omega_r = \sqrt{1 - 2\zeta^2}\,\omega_n \tag{4.35}$$

we have, under ideal conditions (i.e., for $\omega_r = 0$),

$$(\tau_2 + \tau_3)^2 = 2\tau_1 \tau_2 \tag{4.36}$$

The frequency response function of the filter is (see Equation 4.32)

$$G(j\omega) = \frac{\omega_n^2}{\left[\omega_n^2 - \omega^2 + 2j\zeta\omega_n\omega\right]} \tag{4.37}$$

Now for $\omega \ll \omega_n$, , the filter frequency response is flat with a unity gain. For $\omega \gg \omega_n$, the filter frequency response can be approximated by

$$G(j\omega) = -\frac{\omega_n^2}{\omega^2}$$

In a log (magnitude) versus log (frequency) scale, this function is a straight line with slope = –2. Hence, when the frequency increases by a factor of 10 (i.e., one decade), the $\log_{10}$ (magnitude) drops by 2 units (i.e., 40 dB). In other words, the roll-off rate is – 40 dB/decade. Also, ω_n can be taken as the filter cutoff frequency. Hence,

$$\omega_c = \frac{1}{\sqrt{\tau_1 \tau_2}} \tag{4.38}$$

It can be easily verified that when $\zeta = 1/\sqrt{2}$, this frequency is identical to the half-power bandwidth (i.e, the frequency at which the transfer function magnitude becomes $1/\sqrt{2}$).

Note that if two single-pole stages (of the type shown in Figure 4.14(b)) are cascaded, the resulting two-pole filter has an overdamped (nonoscillatory) transfer function, and it is not possible to achieve $\zeta = 1/\sqrt{2}$ as in the present case. Also, note that a three-pole low-pass Butterworth filter can be obtained by cascading the two-pole unit shown in Figure 4.15 with a single-pole unit shown in Figure 4.14(b). Higher order low-pass Butterworth filters can be obtained in a similar manner by cascading an appropriate selection of basic units.

4.4.3 High-Pass Filters

Ideally, a high-pass filter allows through it all signal components above a certain (cutoff) frequency and blocks off all signal components below that frequency. A single-pole high-pass filter is shown in Figure 4.16. As for the low-pass filter that was discussed earlier, the passive filter stage (Figure 4.16(a)) and the active filter stage (Figure 4.16(b)) have identical transfer functions. The active filter is desired, however, because of its many advantages, including negligible loading error due to high input impedance and low output impedance of the op-amp voltage follower that is present in this circuit.

Filter equation is obtained by considering current balance in Figure 4.16(a), noting that the output is in open circuit (zero load current). Accordingly,

$$C\frac{d}{dt}(v_1 - v_o) = \frac{v_o}{R}$$

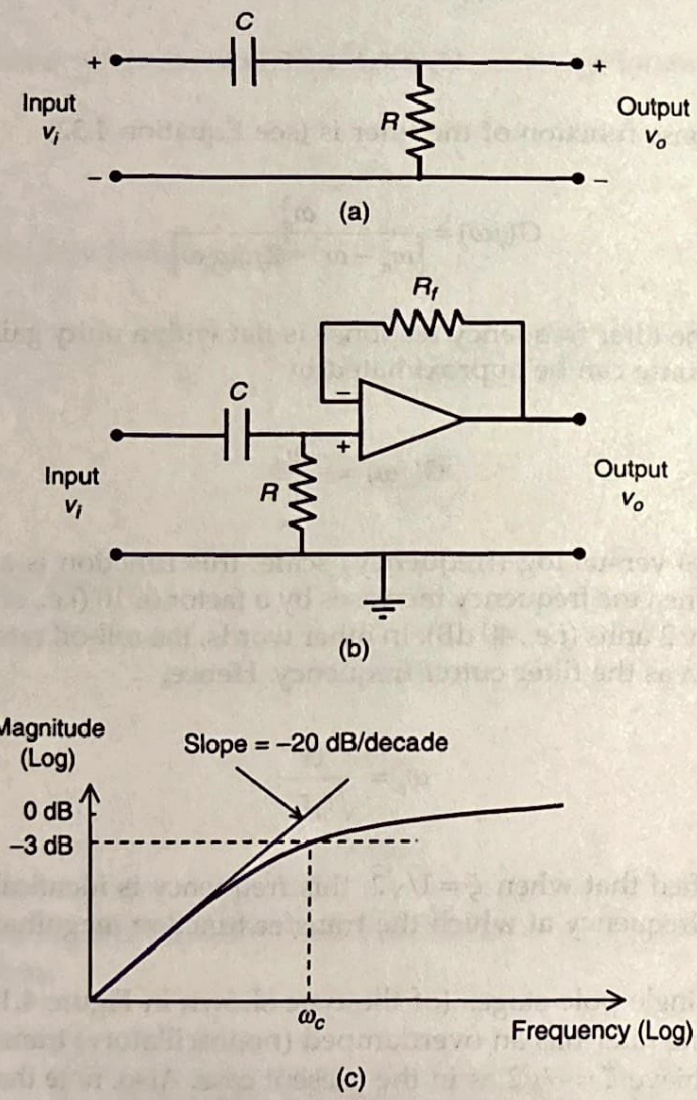


FIGURE 4.16
A single-pole high-pass filter: (a) A passive filter stage, (b) An active filter stage, (c) Frequency response characteristic.

or

$$\tau \frac{dv_o}{dt} + v_o = \tau \frac{dv_i}{dt} \tag{4.39}$$

in which the filter time constant

$$\tau = RC \tag{4.40}$$

Introducing the Laplace variable s, the filter transfer function is obtained as

$$\frac{v_o}{v_i} = G(s) = \frac{\tau s}{(\tau s + 1)} \tag{4.41}$$

Note that this corresponds to a "lead circuit" (i.e., an overall phase lead is provided by this transfer function), as discussed in Chapter 12. The frequency response function is

$$G(j\omega) = \frac{\tau j\omega}{(\tau j\omega + 1)} \tag{4.42}$$

Since its magnitude is zero for $\omega \ll 1/\tau$ and it is unity for $\omega \gg 1/\tau$ we have the cutoff frequency

$$\omega_c = \frac{1}{\tau} \tag{4.43}$$

Signals above this cutoff frequency should be allowed undistorted, by an ideal high-pass filter, and signals below the cutoff should be completely blocked off. The actual behavior of the basic high-pass filter discussed above is not that perfect, as observed from the frequency response characteristic shown in Figure 4.16(c). It can be easily verified that the half-power bandwidth of the basic high-pass filter is equal to the cutoff frequency given by Equation 4.43, as in the case of the basic low-pass filter. The roll-up slope of the single-pole high-pass filter is 20 dB/decade. Steeper slopes are desirable. Multiple-pole, high-pass, Butterworth filters can be constructed to give steeper roll-up slopes and reasonably flat pass-band magnitude characteristics.

4.4.4 Band-Pass Filters

An ideal band-pass filter passes all signal components within a finite frequency band and blocks off all signal components outside that band. The lower frequency limit of the pass band is called the *lower cutoff frequency* (ω_{c1}) and the upper frequency limit of the band is called the *upper cutoff frequency* (ω_{c2}). The most straightforward way to form a band-pass filter is to cascade a high-pass filter of cutoff frequency ω_{c1} with a low-pass filter of cutoff frequency ω_{c2}. Such an arrangement is shown in Figure 4.17. The passive circuit shown in Figure 4.17(a) is obtained by connecting together the circuits shown in Figure 4.14(a) and Figure 4.16(a). The active circuit shown in Figure 4.17(b) is obtained by connecting a voltage follower op-amp circuit to the original passive circuit. Passive and active filters have the same transfer function, assuming that loading problems (component interaction) are not present in the passive filter. Since loading errors and interactions can be serious in practice, however, the active version is preferred.

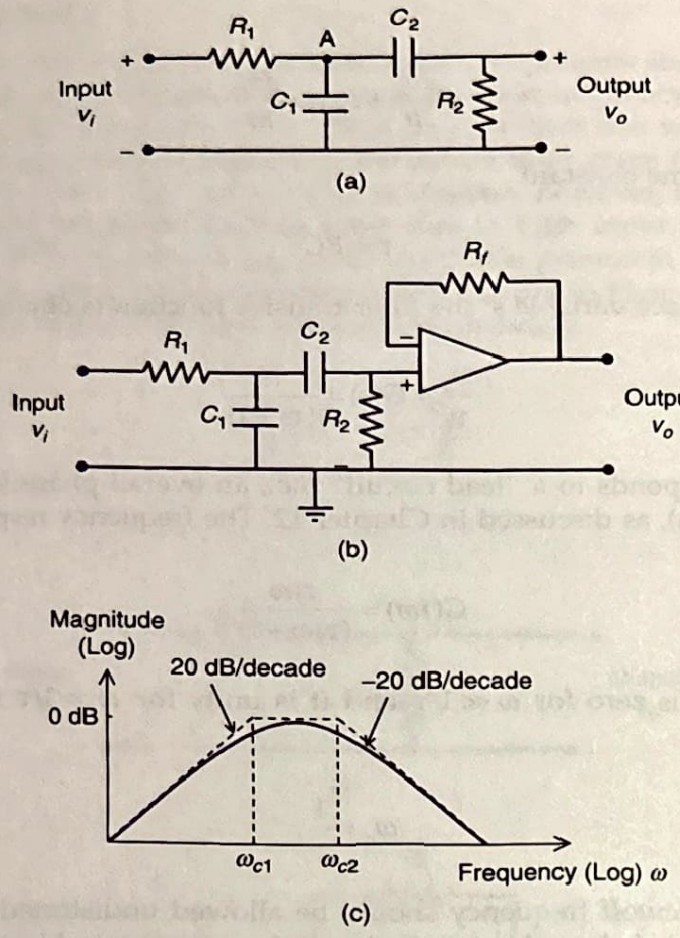


FIGURE 4.17
Band-pass filter: (a) A basic passive filter stage, (b) A basic active filter stage, (c) Frequency response characteristic.

To obtain the filter equation, first consider the high-pass portion of the circuit shown in Figure 4.17(a). Since the output is in open-circuit (zero current) we have from Equation 4.41

$$\frac{v_o}{v_A} = \frac{\tau_2 s}{(\tau_2 s + 1)} \tag{i}$$

in which

$$\tau_2 = R_2 C_2 \tag{4.44}$$

Next, on writing the current balance at node A of the circuit we have

$$\frac{v_i - v_A}{R_1} = C_1 \frac{dv_A}{dt} + C_2 \frac{d}{dt}(v_A - v_o) \tag{ii}$$

Introducing the Laplace variable s (see Appendix A) we get

$$v_i = (\tau_1 s + \tau_3 s + 1) v_A - \tau_3 s v_o \tag{iii}$$

in which

$$\tau_1 = R_1 C_1 \tag{4.45}$$

and

$$\tau_3 = R_1 C_2 \tag{4.46}$$

Now on eliminating v_A by substituting Equation i in Equation iii we get the band-pass filter transfer function:

$$\frac{v_o}{v_i} = G(s) = \frac{\tau_2 s}{\left[\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + \tau_3)s + 1\right]} \tag{4.47}$$

We can show that the roots of the characteristic equation

$$\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + \tau_3)s + 1 = 0 \tag{4.48}$$

are real and negatives. The two roots are denoted by $-\omega_{c1}$ and $-\omega_{c2}$ and they provide the two cutoff frequencies shown in Figure 4.17(c). It can be verified that, for this basic band-pass filter, the roll-up slope is +20 dB/decade and the roll-down slope is –20 dB/decade. These slope magnitudes are not sufficient in many applications. Furthermore, the flatness of the frequency response within the pass band of the basic filter is not adequate as well. More complex (higher order) band-pass filters with sharper cutoffs and flatter pass bands are commercially available.

4.4.4.1 *Resonance-Type Band-Pass Filters*

There are many applications where a filter with a very narrow pass band is required. The tracking filter mentioned in the beginning of the section on analog filters is one such application. A filter circuit with a sharp resonance can serve as a narrow-band filter. Note that the cascaded *RC* circuit shown in Figure 4.17 does not provide an oscillatory response (filter poles are all real) and, hence, it does not form a resonance-type filter. A slight modification to this circuit using an additional resistor R_1 as shown in Figure 4.18(a) will produce the desired effect.

To obtain the filter equation, note that for the voltage follower unit

$$v_A = v_o \tag{i}$$

Next, since current through an op-amp lead is zero, for the high-pass circuit unit (see Equation 4.41), we have

$$\frac{v_A}{v_B} = \frac{\tau_2 s}{(\tau_2 s + 1)} \tag{ii}$$

in which

$$\tau_2 = R_2 C_2$$

Finally, the current balance at node B gives

$$\frac{v_i - v_B}{R_1} = C_1 \frac{dv_B}{dt} + C_2 \frac{d}{dt}(v_B - v_A) + \frac{v_B - v_o}{R_1}$$

or, by using the Laplace variable, we get

$$v_i = (\tau_1 s + \tau_3 s + 2) v_B - \tau_3 s v_A - v_o \tag{iii}$$

Now, by eliminating v_A and v_B in Equations i through Equation iii we get the filter transfer function

$$\frac{v_o}{v_i} = G(s) = \frac{\tau_2 s}{\left[\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + \tau_3)s + 2\right]} \tag{4.49}$$

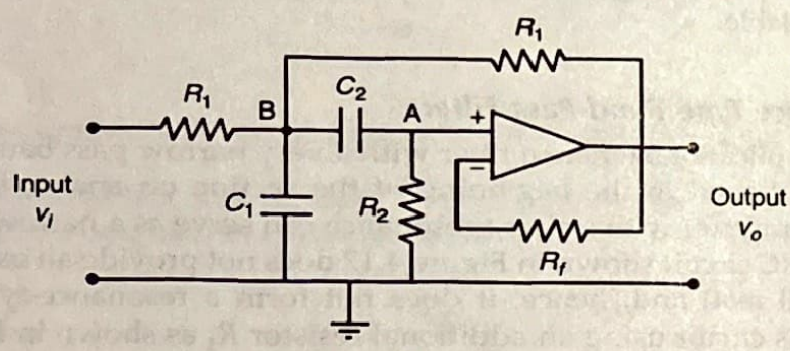
It can be shown that, unlike Equation 4.48, the present characteristic equation

$$\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + \tau_3)s + 2 = 0 \tag{4.50}$$

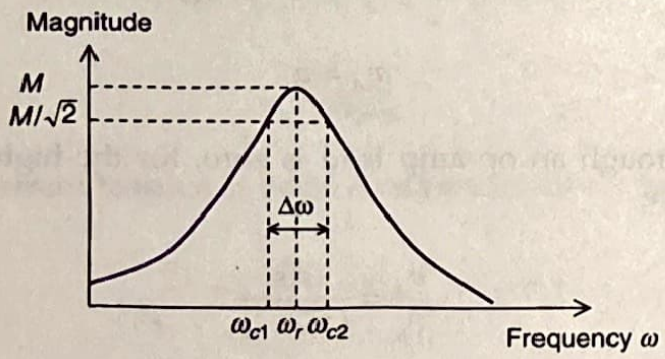
can possess complex roots.

Example 4.8

Verify that the band-pass filter shown in Figure 4.18(a) can have a frequency response with a resonant peak as shown in Figure 4.18(b). Verify that the half-power bandwidth $\Delta\omega$ of the filter is given by $2\zeta\omega_r$ at low damping values. (Note: ζ = damping ratio and ω_r = resonant frequency).



(a)



(b)

FIGURE 4.18
A resonance-type narrow band-pass filter: (a) An active filter stage, (b) Frequency response characteristic.

SOLUTION

We may verify that the transfer function given by Equation 4.49 can have a resonant peak by showing that the characteristic Equation 4.50 can have complex roots. For example, if we use parameter values $C_1 = 2$, $C_2 = 1$, $R_1 = 1$, and $R_2 = 2$ we have $\tau_1 = 2$, $\tau_2 = 2$, and $\tau_3 = 1$. The corresponding characteristic equation is

$$4s^2 + 5s + 2 = 0$$

which has the roots

$$-\frac{5}{8} \pm j\frac{\sqrt{7}}{8}$$

and are obviously complex.

To obtain an expression for the half-power bandwidth of the filter, note that the filter transfer function may be written as

$$G(s) = \frac{ks}{\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)} \tag{4.51}$$

in which

ω_n = undamped natural frequency
ζ = damping ratio
k = a gain parameter

The frequency response function is given by

$$G(j\omega) = \frac{kj\omega}{\left[\omega_n^2 - \omega^2 + 2j\zeta\omega_n\omega\right]} \tag{4.52}$$

For low damping, resonant frequency $\omega_r \cong \omega_n$. The corresponding peak magnitude M is obtained by substituting $\omega = \omega_n$ in Equation 4.52 and taking the transfer function magnitude; thus,

$$M = \frac{k}{2\zeta\omega_n} \tag{4.53}$$

At half-power frequencies we have

$$\left|G(j\omega)\right| = \frac{M}{\sqrt{2}}$$

or

$$\frac{k\omega}{\sqrt{\left(\omega_n^2 - \omega^2\right)^2 + 4\zeta^2\omega_n^2\omega^2}} = \frac{k}{2\sqrt{2}\zeta\omega_n}$$

This gives

$$\left(\omega_n^2 - \omega^2\right)^2 = 4\zeta^2\omega_n^2\omega^2 \tag{4.54}$$

the positive roots of which provide the pass band frequencies ω_{c1} and ω_{c2}. Note that the roots are given by

$$\omega_n^2 - \omega^2 = \pm 2\zeta\omega_n\omega$$

Hence, the two roots ω_{c1} and ω_{c2} satisfy the following two equations:

$$\omega_{c1}^2 + 2\zeta\omega_n\omega_{c1} - \omega_n^2 = 0$$

$$\omega_{c2}^2 - 2\zeta\omega_n\omega_{c2} - \omega_n^2 = 0$$

Accordingly, by solving these two quadratic equations and selecting the appropriate sign, we get

$$\omega_{c1} = -\zeta\omega_n + \sqrt{\omega_n^2 + \zeta^2\omega_n^2} \tag{4.55}$$

$$\omega_{c2} = \zeta\omega_n + \sqrt{\omega_n^2 + \zeta^2\omega_n^2} \tag{4.56}$$

The half-power bandwidth is

$$\Delta\omega = \omega_{c2} - \omega_{c1} = 2\zeta\omega_n \tag{4.57}$$

Now, since $\omega_n \cong \omega_r$ for low ζ we have

$$\Delta\omega = 2\zeta\omega_r \tag{4.58}$$

This result is identical to what was reported in Chapter 2.

A notable shortcoming of a resonance-type filter is that the frequency response within the bandwidth (pass band) is not flat. Hence, quite nonuniform signal attenuation takes place inside the pass band.

4.4.5 Band-Reject Filters

Band-reject filters or *notch filters* are commonly used to filter out a narrow band of noise components from a signal. For example, 60 Hz line noise in a signal can be eliminated by using a notch filter with a notch frequency of 60 Hz.

An active circuit that could serve as a notch-filter is shown in Figure 4.19(a). This is known as the Twin T circuit because its geometric configuration resembles two T-shaped circuits connected together. To obtain the filter equation, note that the voltage at point P is v_o because of unity gain of the voltage follower. Now, we write the current balance at nodes A and B; thus,

$$\frac{v_i - v_B}{R} = 2C\frac{dv_B}{dt} + \frac{v_B - v_o}{R}$$

$$C\frac{d}{dt}(v_i - v_A) = \frac{v_A}{R/2} + C\frac{d}{dt}(v_A - v_o)$$

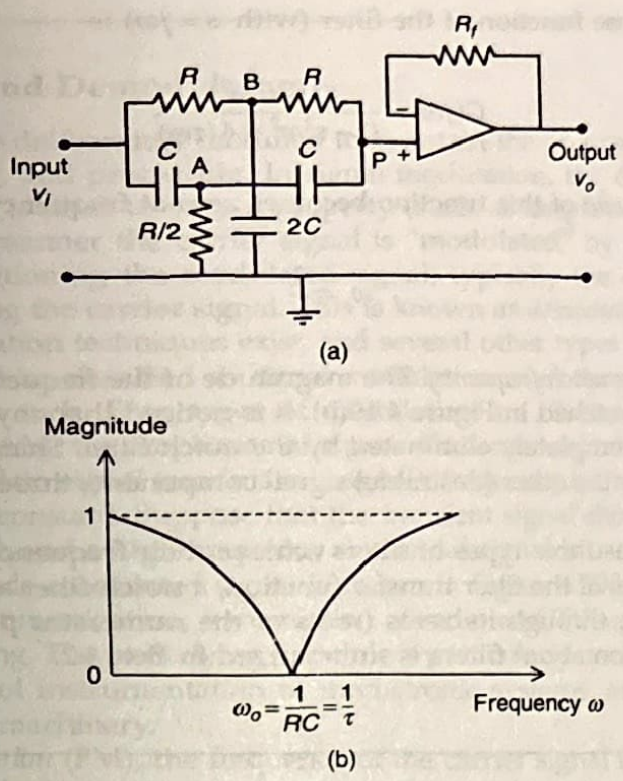


FIGURE 4.19
A notch filter: (a) An active twin T filter circuit, (b) Frequency response characteristic.

Next, since the current through the + lead of the op-amp (voltage follower) is zero, we have the current through point P as

$$\frac{v_B - v_o}{R} = C\frac{d}{dt}(v_o - v_A)$$

These three equations are written in the Laplace form:

$$v_i = 2(\tau s + 1)v_B - v_o \qquad \text{(i)}$$

$$\tau s v_i = 2(\tau s + 1)v_A - \tau s v_o \qquad \text{(ii)}$$

$$v_B = (\tau s + 1)v_o - \tau s v_A \qquad \text{(iii)}$$

in which

$$\tau = RC \qquad (4.59)$$

Finally, eliminating v_A and v_B in Equations i through iii we get

$$\frac{v_o}{v_i} = G(s) = \frac{(\tau^2 s^2 + 1)}{(\tau^2 s^2 + 4\tau s + 1)} \qquad (4.60)$$

The frequency response function of the filter (with $s = j\omega$) is

$$G(j\omega) = \frac{(1-\tau^2\omega^2)}{(1-\tau^2\omega^2 + 4j\tau\omega)} \tag{4.61}$$

Note that the magnitude of this function becomes zero at frequency

$$\omega_o = \frac{1}{\tau} \tag{4.62}$$

This is known as the *notch frequency*. The magnitude of the frequency response function of the notch filter is sketched in Figure 4.19(b). It is noticed that any signal component at frequency ω_o will be completely eliminated by the notch filter. Sharp roll-down and roll-up are needed to allow the other (desirable) signal components through without too much attenuation.

Whereas the previous three types of filters achieve their frequency response characteristics through the poles of the filter transfer function, a notch filter achieves its frequency response characteristic through its zeros (roots of the numerator polynomial equation). Some useful information about filters is summarized in Box 4.2.

BOX 4.2

Filters

Active Filters (Need External Power)

Advantages:

- Smaller loading errors and interaction (have high input impedance and low output impedance, and hence don't affect the input circuit conditions, output signals and other components).
- Better accuracy

Passive Filters (No External Power, Use Passive Elements)

Advantages:

- Useable at very high frequencies (e.g., radio frequency)
- No need of a power supply
- Lower cost

Filter Types

- Low Pass: Allows frequency components up to cutoff and rejects the higher frequency components.
- High Pass: Rejects frequency components up to cutoff and allows the higher frequency components.
- Band Pass: Allows frequency components within an interval and rejects the rest.
- Notch (or, Band Reject): Rejects frequency components within an interval (usually, a narrow band) and allows the rest.

Definitions

- Filter Order: Number of poles in the filter circuit or transfer function
- Anti-aliasing Filter: Low-pass filter with cutoff at less than half the sampling rate (i.e., at less than Nyquist frequency), for digital processing.
- Butterworth Filter: A high-order filter with a quite flat pass band.
- Chebyshev Filter: An optimal filter with uniform ripples in the pass band
- Sallen-Key Filter: An active filter whose output is in phase with input.